CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

April 2013

CL 306 Transportation Engineering-II

Date: 30.04.2013, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Do as directed: **[07]**

(a) The gradients of railway tracks are flatter than roadways- True or False?

(b) Indian railways are divided into ________ zones. (09,16,04)

(c) Name two navigational aids of ports.

(d) Name two storage facilities at port.

(e) Name two loading and unloading facilities at port.

(f) What is Ferry?

(g) Name two types of entrances for ship at harbor.

Q - 2 (a) I. What are the types of gradients used on the railway? **[02]**

II. Why cant deficiency is limited to 75 mm on B.G.? **[02]**

III. The maximum sanctioned speed of the section is 100 km/h. The equilibrium speed may be taken as 75% of the MSS. Calculate superelevation for a 1° curve. **[02]**

IV. The railway board has decided that on Indian railways transition curves will normally be laid in the shape of ____________. **[01]**
(cubic parabola, cubic spiral, lemniscates)

V. Write equation to find degree of a curve. **[01]**

(b) What are the problems due to different gauge? **[06]**

OR

Q - 2 (a) Discuss about cant deficiency and cant excess for design of railway track. **[04]**

(b) Discuss about necessity and various parameters of geometric design of track. **[04]**

(c) I. Min. cushion of __________ of ballast below sleeper is provided in Indian railways? **[01]**
(10 cm, 20 cm, 30 cm)

II. Size of ballast universally adopted for all types of sleeper is ________. [01]
(25 mm, 50 mm, 80 mm)

III. Discuss – black cotton soil is not suitable as sub grade for railway track. [03]

IV. Write function of fish plate. [01]

Q - 3 (a) Define switch and crossing. [02]

(b) Discuss about classification of stations and yards. [08]

(c) Discuss about semaphore signal in brief. [04]

OR

Q - 3 (a) Enlist types of crossing. [02]

(b) List out facilities provided at railway station and equipments at yards. [08]

(c) Discuss about mechanical method of interlocking in brief. [04]

SECTION – II

Q - 4 (a) Apply the corrections for elevation, for temperature and for gradient only and determine the runway length to be provided from the following data: [03]

(i) runway length required for landing and takeoff at sea level in standard atmospheric conditions = 2500 m

(ii) airport elevation = 150 m

(iii)airport references temperature = 24^0C

(iv)temperature in standard atmosphere at 150 m = 14.025^0C

(v) runway slope = 0.5%

(b) Give the recommended phases of master plan according to FAA. [04]

Q - 5 (a) Enlist characteristics of good harbor. [04]

(b) Define dolphins; give classification and function of dolphin with neat sketch. [05]

(c) Write a note on minimum turning radius with neat sketch. [05]

OR

Q - 5 (a) Define: (i) Roadstead, (ii) Littoral drift and draw a neat sketch of artificial circumscribed roadstead. [04]

(b) Define breakwater; write a note on rubble mound breakwater with neat sketch. [05]

(c) Define: (i) pay load, (ii) maximum gross takeoff weight and write a note on minimum circling radius [05]

Q - 6 (a) Enlist functions of AAI. [03]

(b) Enlist site selection criteria for airport and explain development of surrounding area. [05]

(c) Define basic runway length and briefly describe the classification based on landing and takeoff of basic runway length with neat sketch. [06]

OR

Q - 6 (a) Give a classification of airports. [03]

(b) Explain in detail atmospheric and meteorological conditions of selection criteria of an airport. [05]

(c) The length of runway at sea level under standard condition at zero gradient is 1500 m. the airport site is at an elevation of 900 m, the reference temperature is $20^{0}C$ and the proposed runway permits ascending gradient of 0.5%. Design the longitudinal profile of runway. [06]

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